Continual Learning With Structured Inheritance for Semantic Segmentation in Aerial Imagery

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***Abstract*— With the rapid update and iteration of current aerial image data, the continual learning scenarios and catastrophic forgetting problem attracted increased attention, especially in the semantic segmentation task. However, the existing methods mainly focus on the class continual learning in a single task and are not satisfactory when extended to multiple tasks. In this article, we consider more realistic and complicated settings, namely task continual learning. We revisit the characteristics of semantic segmentation and knowledge distillation (KD) strategy, then propose a general and effective framework, named structured inheritance, to learn new tasks while retaining high performance on old tasks. Specifically, we present two structure-preserving penalties: pixel affinity structure loss and representation consistency structure loss. The former breaks the isolation of pixels and retains the pixel interactive information learned by the old tasks. At the same time, the latter protects high-frequency stationary information between sequence semantic segmentation tasks. Our approach does not need to add extra parameters nor does it need to access the data stream of the old tasks. Therefore, it can be applied in practical applications with strict computational burden, memory cost, and storage budget. Extensive continual learning experiments on four semantic segmentation datasets of Vaihingen, Potsdam, DeepGlobe, and Gaofen challenge semantic segmentation dataset (GCSS) prove the effectiveness of our proposed framework, which outperforms the current state-of-the-art methods and even exceeds the theoretical upper-bound performance of multitask learning. The code and models will be made publicly available.**

***Index Terms*— Continual learning, deep convolution neural network, remote sensing, semantic segmentation.**

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I. Introduction

With the development of Earth observation technology, the acquisition capability and iteration speed of aerial images have been improved while bringing serious challenges to the current remote sensing semantic segmentation methods. The most of existing methods [1]–[7] are typically trained offline and deployed statically. When encountering new tasks/datasets, they will suffer catastrophic forgetting problem and has to discard the previous knowledge to retrain the model [8]. In a wide range of remote sensing applications, such as satellite on-orbit interpretation and unmanned aerial vehicle (UAV) real-time monitoring, the strategy of retraining the model is costly or even infeasible. Therefore, there is an urgent need to ensure that the model has the ability to learn new tasks while retaining high performance on old tasks. In this case, continual learning [9] is an effective strategy to solve the above problems, which has received increased attention.

According to the statistical learning theory, the cause of the catastrophic forgetting problem is that different tasks or datasets are independent but not identically distributed (i.n.i.d.) in the feature domain. As shown in Fig. 1, the distributions of different remote sensing datasets vary greatly. They contain intricate variations in resolution, image size, color textures, semantic labels, and imaging methods. Therefore, the essence of continual learning to solve the catastrophic forgetting problem is to generalize different feature distributions without accessing historical data.

More recently, the continual learning paradigm has been extensively studied. The existing methods can be divided into three families: network expansion [10]–[17], parameter isolation [18]–[22], and regularization [23]–[30] methods. Network expansion methods propose to freeze existing parameters and dynamically add parameters or subnets for each new task. Parameter isolation methods assume that the parameters are independent of each other. They explore various strategies to measure the importance of each parameter to old tasks and penalize the changes of important parameters when training on the new tasks. Besides, regularization methods utilize the information hidden in the old model to retain the historical knowledge of the old tasks based on the knowledge distillation (KD) strategy [31].

These methods have been validated in image classification (CLS) and class continual learning setting (all classes are from the same dataset), especially regularization methods. However,

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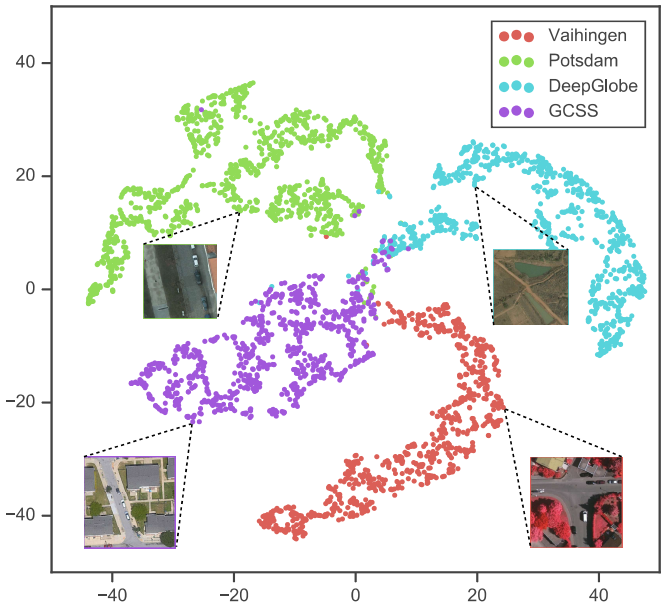


Fig. 1. t-stochastic neighbor embedding (SNE) visualization results of four semantic segmentation datasets, Vaihingen, Potsdam, DeepGlobe, and GCSS. We can find that the samples from different datasets are independent nonidentically distributed.

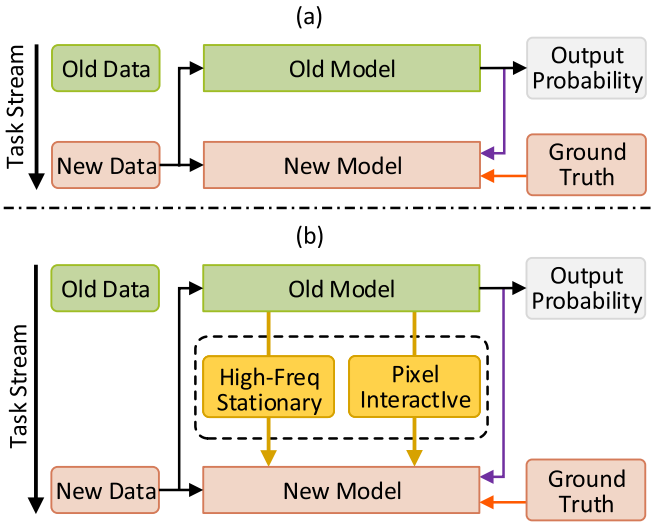
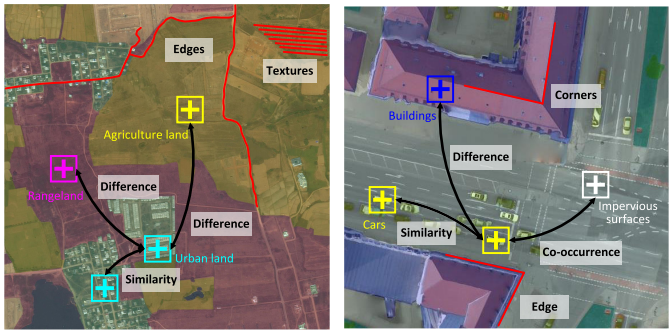


Fig. 2. Schematic of the continual learning pipeline. (a) In the previous works, they only consider the output probability to avoid catastrophic forgetting problem. It is not sufficient in complex semantic segmentation tasks. (b) In our method, we propose to retain pixel interactive information and protect high-frequency stationary information.

as shown in Fig. 2, we find that previous works [32]–[35] show limited performance when they are directly applied to task continual learning (all tasks belong to different datasets) of remote sensing semantic segmentation. The main reasons as follows: one is that there is no retention of pixel interactive information. The remote sensing images usually have a much larger size. For example, the images collected by the Beijing-2 satellite have $47\,244 \times 47\,244$ pixels [2]. Moreover, remote sensing images are mostly composed of amorphous areas, such as water, forest, and vegetation. Therefore, as shown in Fig. 3, the pixel interactive information, such as cooccurrence and



(a) (b)

Fig. 3. Illustration of pixel interactive information (black arrow) and high-frequency stationary information (red line). (a) Sample with ground truth from DeepGlobe. (b) Sample with ground truth from Potsdam. We can see that "Urban land" is different from Agriculture land, and "road" and "car" appear together with a high probability. Moreover, the high-frequency stationary property, like edges, textures, corners, will not be affected by categories and datasets. (Best viewed in color.)

similarity or difference, plays an essential role in improving segmentation performance, let alone avoiding catastrophic forgetting on old remote sensing tasks. The other is that there is no protection for high-frequency stationary information. The remote sensing images have large amounts of data and complicated backgrounds, which increases the difficulty of optimizing the semantic segmentation model. Therefore, the model is more inclined to learn high-level features with rich semantics and strong task relevance, making the high-frequency stationary property easy to be overwhelmed. However, as shown in Fig. 3, the task-agnostic high-frequency stationary property, such as the corners of buildings and the edges of roads, is similar or even unchanged in different remote sensing tasks/datasets, which is more conducive to the modeling of generality between tasks.

In response to the above problem, we revisit the characteristics of semantic segmentation and KD strategy. Then, we propose the task continual learning framework tailored to semantic segmentation in aerial imagery based on the idea of structured inheritance. Specifically, it incorporates two structure-preserving penalties to protect the structure information: pixel affinity structure (PAS) loss and representation consistency structure (RCS) loss.

The PAS loss is proposed to retain the pixel interactive information learned from the old tasks. Inspired by the significant performance gain of global context information to semantic segmentation networks [6], [7], we first construct the affinity matrix across the whole image by capturing the long-range dependence of features at spatial positions. Then, the PAS loss constrains the affinity matrices between the old and new models, effectively breaking the isolation of pixels and retaining the pixel interactive information.

The RCS loss is inspired by the interpretability of convolutional networks [36]–[39], the shallow layers with high-resolution learn the generalization and task-agnostic stationary features, such as corners, textures, and edge. Therefore, the RCS loss is used to align the high-frequency embeddings consistencies between the shallow layer of the old and new models. The purpose is to protect the high-frequency stationary information between sequence semantic segmentation tasks.

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To this end, we optimize the objective function that combines the cross entropy loss and output probability distillation (OPD) loss with proposed structure-preserving penalties. We can find that our framework does not need to add additional parameters or access historical data, which can be applied in practical applications with the strict computational burden and resource constraints. In order to verify the generality and effectiveness of our framework, we conduct experiments on four semantic segmentation datasets, including two International Society for Photogrammetry and Remote Sensing (ISPRS) semantic labeling contest dataset [40], DeepGlobe land cover challenge dataset [41], and Gaofen Challenge Semantic Segmentation dataset (GCSS) [42], [43]. Experimental results show that our method is superior to the existing continual learning methods and even exceeds the theoretical upper-bound performance of multitask learning [44], [45]. The contributions of this article can be summarized as follows.

1) We propose an effective and general framework based on structured inheritance, which does not require additional storage requirements and parameters, so it is more suitable for practical applications.
2) We present the PAS loss to constrain the affinity matrices built on the old and new models, which effectively breaks the isolation of pixels and retains the pixel interactive information learned from old tasks.
3) The RCS loss is proposed to align the high-frequency embeddings consistencies between the shallow layer of the old and new models, which can protect the high-frequency stationary information.
4) Experiments with four public semantic segmentation datasets show that our proposed framework is more effective than existing state-of-the-art methods and even exceeds the theoretical upper-bound performance of multitask learning.

II. Related Works

The catastrophic forgetting problem and continual learning paradigm have been extensively studied in many previous works. These approaches can be grouped into three main families, namely network expansion, parameter isolation, and regularization methods. We briefly review recent works as follows.

A. Network Expansion Methods

The network expansion methods [10]–[17] assign different parameter subsets of the model to each task. For, example, progressive neural networks (PNN) [10] keeps the weights fixed and add lateral connections at each layer to learn new tasks. Expert Gate [11] establishes an expert network for each task and learning an autoencoder gate to activate the corresponding expert network according to the input task. Packnet [12] performs iterative pruning and network retraining to identify the parts that are used for each task. Continual neural-Dirichlet process mixture (CN-DPM) [14] proposes a set of neural network experts that are in charge of a subset of the data to solve the task-free continual learning problem. The network overhead of these methods will increase dramatically as the number of tasks increases. Therefore, under a fixed deep neural network (DNN) architecture, Piggyback [17] learns the binary masks to create different subnets for each task. However, network expansion methods are computationally expensive, and the network capacity is quickly exhausted, they are not suitable for large-scale continual learning and cannot be deployed on edge devices with strict limitations on memory and computation.

B. Parameter Isolation Methods

The parameter isolation methods [18]–[22] advocate that the parameters that matter most to the performance for old tasks should remain fixed. Therefore, these methods focus on how parameter importance is calculated. Elastic weight consolidation (EWC) [18] analyzes the continual learning framework from the perspective of sequential Bayesian estimation and proves that the second derivative of the parameters can measure the importance of parameters. Memory aware synapses (MAS) [19] proposes an unsupervised manner to estimate the importance of the weights. Incremental moment matching (IMM) [20] proposes a separate model merging step after learning each new task. Path integral (PI) [21] calculates the importance of each parameter for one task based on the impact of its update on loss reduction. Riemannian walk (RW) [22] combines the PI with EWC to measure the importance of parameters for each task. However, these methods are based on the assumption of parameter independence, which limits the upper bound of the performance, especially in the field of high task complexity.

C. Regularization Methods

The regularization methods [23]–[25] utilize the knowledge of the previous models to mitigate the catastrophic forgetting problem. Learning without forgetting (LWF) [23] first exploits the KD strategy [31] to the continual learning scenarios. Specifically, it uses the images of a new task as the input of the old model and its outputs as the supervision information of the old tasks. Based on LWF, learning without memorizing (LWM) [25] proposes an attention distillation loss to further preserve the information about old tasks. Moreover, some works store samples [26]–[30] or generate high-quality pseudo-samples [46], [47] of old tasks to further alleviate the catastrophic forgetting problem.

Inspired by the effectiveness of regularization methods, some pioneer works have introduced the distillation strategy into other fields, such as object detection [48], [49] and semantic segmentation [32]–[35]. To our knowledge, KD-binary cross entropy (BCE) [32] is the only work trying to solve continual learning of remote sensing semantic segmentation. They extend the method [26] proposed for image CLS to semantic segmentation and devise a strategy to select representative samples of the old classes for replay. However, like the continual learning method of semantic segmentation applied to medical images [50], [51] and general images [33]–[35], KD-BCE considers only the class continual learning setting, that is, both the new class and the old class come from training data with the same distribution. This assumption dramatically limits their applicability. In this work, we do

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not limit our analysis to a single dataset. On the contrary, we consider more realistic and complicated settings where the stream of data can come from multiple semantic segmentation datasets with different distributions and propose to retain the pixel interactive information and high-frequency stationary information of the semantic segmentation tasks.

D. Semantic Segmentation of Aerial Imagery

With the introduction of a fully convolution network (FCN) [52] and its variants [53], [54], the semantic segmentation of aerial images has made great progress. Recently, a number of works [1]–[7] have a focus on the learning of global context information, which further improves the performance of semantic segmentation. For example, Nogueira *et al.* [1] exploit a multicontext paradigm and capture multicontext characteristics from heterogeneous contexts. Class-guided feature decoupling network (CGFDN) [4] encoded the class cooccurrence relations while capturing the spatially contextual information. Mou *et al.* [5] proposed to learn and reason about global relationships between any two spatial positions. However, when encountering sequential learning tasks, these methods will plague with the catastrophic forgetting problem, they can only learn to solve the most recent task, but rapidly forgets about previous tasks. In this article, we aim to explore a more effective way to address task continual learning of remote sensing semantic segmentation. Compared with image CLS and class continual learning setting, it is more realistic but also more challenging.

III. Methodology

In this section, we describe our structured inheritance framework. First, the problem definition of continual learning is given in Section III-A. Section III-B introduces the overall network architecture. Then, we introduce in detail the proposed two structure-preserving penalties in Section III-C and Section III-D. Finally, Section III-E illustrates the training procedure of the proposed framework.

A. Problem Definition

Continual learning studies the problem of learning multiple tasks as a sequential stream of activities, with the goal of overcoming catastrophic forgetting and providing a competitive Multitask classifier for the tasks observed so far [9]. Compared with Multitask learning [44], [45], the training data of continual learning can only be accessed sequentially, the update of the model can only utilize the training data of the current task. Therefore, continual learning is an effective method to solve scenarios with privacy concerns in a realistic environment.

Formally, considering a sequence of supervised learning tasks, denoted by $\mathcal{T} = (T_1, T_2, \ldots, T_t, \ldots)$. Each task T_t takes the form of a semantic segmentation problem and has training data, $\mathcal{D}^t_{\text{train}} = \{x^t_i, y^t_i\}|^{n_t}_{i=1}$, where n_t is the total number of the training data, x^t_i is the ith input image, and $y^t_i \in \{1, 2, \ldots, C_t\}$ is its corresponding ground truth map, where C_t is the total number of classes for task T_t. Similarly, we denote $\mathcal{D}^t_{\text{test}}$ as the test data for task T_t. Therefore, the training and test data for tasks from T_1 to T_t can be denoted as $\mathcal{D}^{1:t}_{\text{train}} = \cup^t_{t=1}\mathcal{D}^t_{\text{train}}$ and $\mathcal{D}^{1:t}_{\text{test}} = \cup^t_{t=1}\mathcal{D}^t_{\text{test}}$, respectively. The set of all learned parameters for the model up to the current task T_t is given as $\Theta_t = \{\theta^k_t\}$, where k means the different layer of the model. The continual learning model requires a reasonable number of parameters and memory, and should not continue to increase as the number of training tasks increases. The training phase of the continual learning is implemented in multiple stages, called learning steps. During the learning step t, the model can only observe the training data $\mathcal{D}^t_{\text{train}}$. In contrast, the previously learned training data $\mathcal{D}^{1:t-1}_{\text{train}}$ are no longer accessible, let alone the knowledge of the relationship of tasks or duration of time. The critical issue is to train the model without catastrophic forgetting on old tasks or significant underperformance on the new task.

B. Network Architecture

Our proposed framework is shown in Fig. 4, we propose a multihead continual learning network. During the learning step t, the network takes the training sample of task T_t as the input of the new model and the old model. The cross entropy loss is used to learn the new task. At the same time, the OPD loss and proposed structure-preserving penalties are used to avoid the degradation of the accuracy of the old tasks. The continual learning network consisting of the task-shared layers and task-specific layers, the task-shared layers are used to learn high-dimensional semantic representations from the input data, and the task-specific layers are used to predict the semantic class label. We adopt the PSPNet [53] with ResNet [55] as our task shared layers. The L_1 to L_4 mean the four residual blocks, and PPM means the pyramid pooling module. The task-specific layers are composed of the same number of single 1×1 convolution layer head classifiers as the number of tasks. Considering the computational burden, memory cost, and storage budget in practical applications, we fix the number of parameters and structure of the task shared layers. When training one new task, the number of the added parameters is the number of categories for the new task multiplied by the number of channels at the end of the task shared layers, which is just a small fraction of the total number of parameters. Therefore, we have stricter constraints on the proposed framework, which can meet the needs of actual application scenarios.

During the learning step t, an ideal way to help the continual learning network remember all the observed tasks T_1 to T_t is to minimize the following objective function:

$$\mathcal{L}\left(\Theta_t, \mathcal{D}^{1:t}_{\text{train}}\right) = \sum_{j=1}^{t} \mathcal{L}_j\left(\Theta_j, \mathcal{D}^j_{\text{train}}\right) \tag{1}$$

$$\mathcal{L}_j\left(\Theta_j, \mathcal{D}^j_{\text{train}}\right) = \frac{1}{n_j}\sum_{i=1}^{n_j} \ell_j\left(x^j_i, y^j_i\right) \tag{2}$$

where ℓ_j is the loss for task T_j. In semantic segmentation task, we generally use the cross entropy loss. For task T_t, its defined as follows:

$$\mathcal{L}_{\text{ce}}\left(x^t_i, y^t_i\right) = -\sum_{c \in C_t} y^t_i[c] \cdot \log\left(f\left(x^t_i; \Theta_t\right)[c]\right) \tag{3}$$

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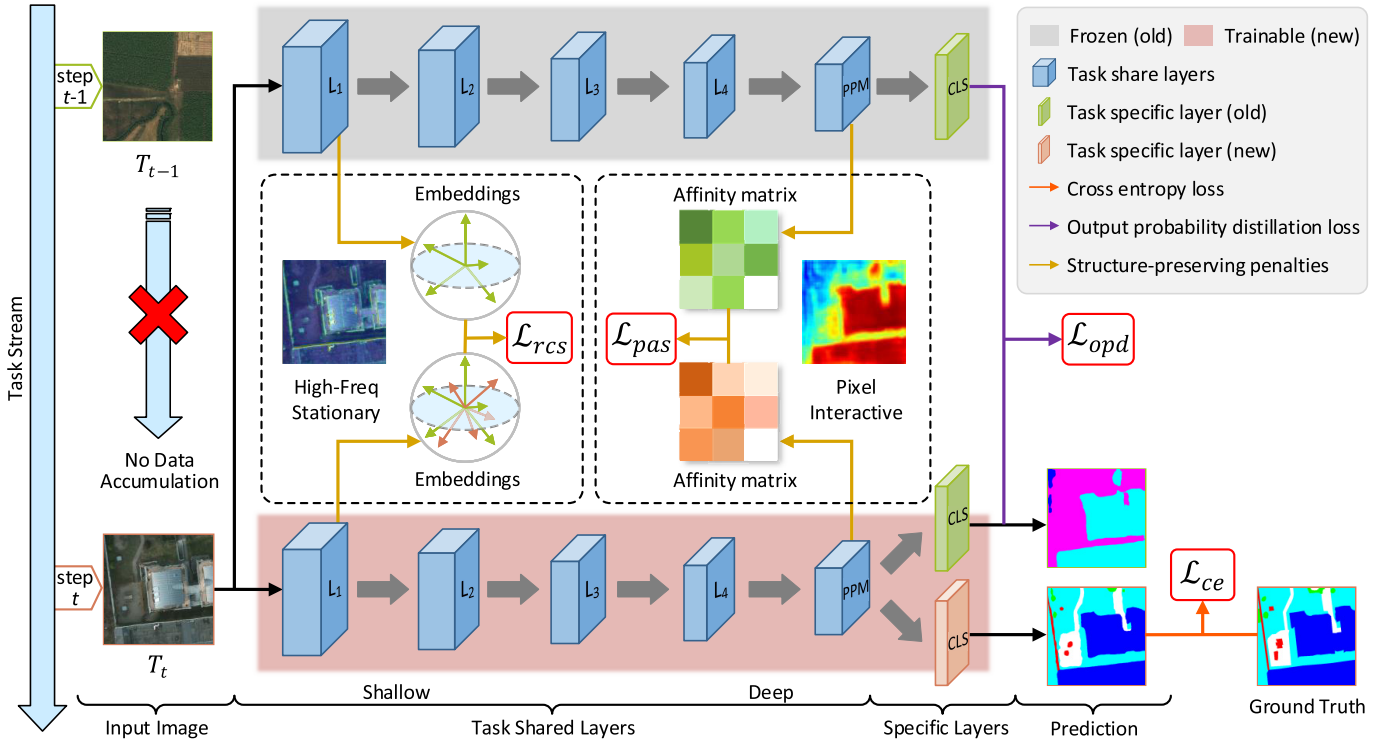

Fig. 4. Pipeline of our framework. The continual learning network consists of the task-shared layers and task-specific layers. The L_1 to L_4 denote the residual blocks, PPM denotes the pyramid pooling module and CLS denotes the head classifier. During the learning step t, the new model Θ_t is completely trainable, while the old model Θ_{t-1} is frozen. The training images of new task are fed into the old and new models to obtain the corresponding mapping outputs, the cross entropy loss ($\mathcal{L}_{\text{ce}}$) is applied to learn new task (light red arrow), the two structure-preserving penalties ($\mathcal{L}_{\text{pas}}$ and $\mathcal{L}_{\text{rcs}}$, orange arrow) and OPD loss ($\mathcal{L}_{\text{opd}}$, purple arrow) are applied to preserve old tasks. Note that we do not store data learned in previous steps. (Best viewed in color.)

where $f(\cdot;\Theta_t)$ means the prediction of the segmentation network for the input image, $y_i^t[c]$ and $f(x_i^t;\Theta_t)[c]$ are one-hot ground truth and the output probability for class c, respectively.

However, the objective function 1 requires entire training data. In continual learning setting, all the previous training data $\mathcal{D}_{\text{train}}^{1:t-1}$ are not available at the current learning step t. Only the model parameters Θ_{t-1} of the last learning step are accessible. Therefore, an intuitive idea is to utilize the model Θ_{t-1} instead of the previous training data. Specifically, the output probabilities of the new model on the old task branch should be close to the output probabilities of the corresponding task branch of the old model. The OPD loss can be defined as follows:

$$\mathcal{L}_{\text{opd}}\left(x_i^t\right) = -\sum_{c\in C_{1:t-1}} \pi_c\left(f\left(x_i^t;\Theta_{t-1}\right)[c]\right) \cdot \log\left(\pi_c\left(f\left(x_i^t;\Theta_t\right)[c]\right)\right) \quad (4)$$

here $C_{1:t-1}=\{C_1, C_2, \ldots, C_{t-1}\}$ denotes all classes of the old tasks, $\pi_c(v) = v_c^{1/\tau}/\sum_j v_j^{1/\tau}$ is the rescaling function, where τ is the temperature parameter to soften and smooth the output probability value.

Although the OPD loss effectively alleviates the catastrophic forgetting problem, we experimentally demonstrate that a naive application of the distillation loss would not sufficient. We find that the important pixel interactive information and high-frequency stationary information of semantic segmentation are neglected. Therefore, we put forward two structure-preserving penalties, i.e., PAS loss and RCS loss, to retain pixel interactive information and high-frequency stationary information without prejudice to the new tasks.

C. PAS Loss

A semantic segmentation task assigns each pixel in an image to a semantic label. It seems like an upgraded version of image CLS. However, it is impossible to simply take the semantic segmentation task as a collection of pixel CLS tasks. The pixel interactive information, such as cooccurrence, similarity or difference, will inevitably affect the final prediction results. In other words, the features learned by the semantic segmentation network must have modeled pixel interactive information to assist its prediction. The state-of-the-art methods proposed by strengthening pixel interactive information have proved its importance [1], [3], [3]–[7]. However, the OPD loss can only learn pixel-wise information. Inspired by this, we propose to break the isolation of pixels and retain the pixel interactive information learned from old tasks.

Based on the above analysis, we first need to calculate the pixel interactive information on the old model, then let the new model learn the information. Therefore, we define the affinity matrix at the end of the shared layer to capture the long-range dependence of features at spatial positions. As shown in Fig. 5(a), given a 3-D tensor outputted by convolutional layers, the affinity matrix can model pixel interactive information of the holistic tensor for the position-wise feature. The affinity matrix has an exact physical meaning, each row of

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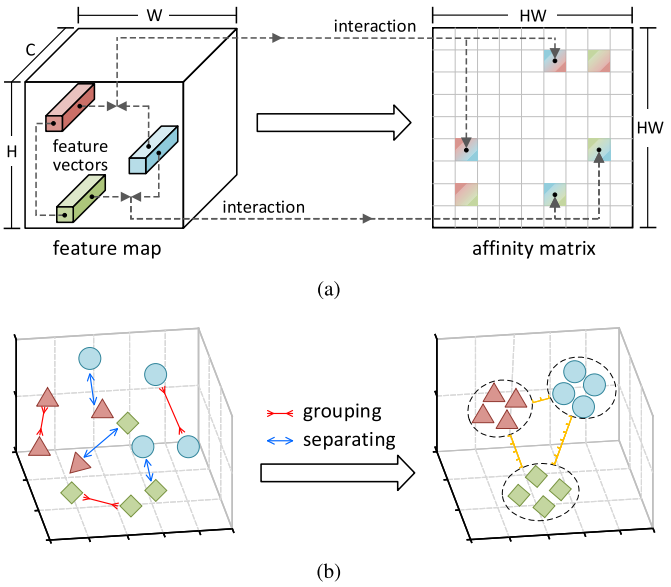

Fig. 5. Illustration of PAS loss. (a) Affinity matrix, which can capture the long-term dependence of the element in the spatial position. (b) Schematic of the effect of PAS loss, which protects the pixel interactive information learned by the old tasks. The pixel of the same class have a strong correlation, and pixels of different classes have different degrees of correlations. (Best viewed in color.)

which indicates the statistical correlation of the corresponding pixel with all pixels.

Formally, assume that the 3-D feature map $F \in R^{C\times H\times W}$ output by the end of the task shared layers $\theta^e(\cdot)$, where W and H mean the spatial width and height, C means the number of channels. For better illustration, we denote the feature map to $R^{C\times N}$, where $N = H \times W$ means the number of pixel feature vectors, so the feature vector of each pixel is denoted as $f^{\cdot k} \in R^C, k = [0, N)$. After that, the affinity matrix $A \in R^{N\times N}$ is calculated by matrix multiplication as follows:

$$A_{ij} = \mathcal{N}\left(f^{\cdot i}\right)^T \cdot \mathcal{N}\left(f^{\cdot j}\right) \tag{5}$$

where $\mathcal{N}(v) = v/\|v\|_2$ denotes the l_2-normalized vector, i and j are the indexes for the pixels. The values of the affinity matrix are relative and has no direct relationship with the category and source (task) of the image pixels. Therefore, the pixel interactive information can be preserved by learning the affinity matrix constructed based on the old model. During the learning step t, the PAS loss can be defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{pas}}\left(x_i^t\right) &= \mathcal{L}_2\left(A\left(\theta_{t-1}^e\left(x_i^t\right)\right), A\left(\theta_t^e\left(x_i^t\right)\right)\right) \\ &= \frac{1}{N^2}\left\|A(F_{t-1}) - A(F_t)\right\|_2^2\end{aligned} \tag{6}$$

where F_{t-1} and F_t denote the feature maps output by the end of the shared layer of old model Θ_{t-1} and new model Θ_t, respectively.

Although the learned training data of old tasks are not accessible, the old model can be used to model the pixel interaction information of the old tasks. As shown in Fig. 5(b), the model can preserve the long-range pixel dependence knowledge of the old tasks through the PAS loss. The interactions of pixels belonging to the same class are strengthened, and pixels of different classes have different degrees of correlations, too. It breaks the isolation of pixels in one image, making up a whole with logic relation to each other, which is important for semantic segmentation. Therefore, it can further constrain the parameter update of the continual learning network to avoid catastrophic forgetting. The qualitative analysis of pixel interactive information can be seen in Section IV-E1.

D. RCS Loss

According to the interpretability of convolutional networks, the convolution networks can learn the low-level and high-level features progressively of the input image [1], [36]–[39]. The deep layers with low-resolution show significant variation, capturing more task-specific and semantically stronger features, the shallow layers with high-resolution learn the generalization and task-agnostic low-lever features, that is, high-frequency stationary information of the image, such as corners, textures, and edge, etc. Inspired by this, to avoid the catastrophic forgetting problem, the high-frequency stationary information of the shallow layer should not change drastically as the number of tasks increases. That is to say, the high-frequency information of the shallow layer on different learning steps should be generality. Therefore, we propose to protect the high-frequency stationary information while updating the weights of the continual learning network.

The most straightforward strategy to achieve this goal is to freeze weights or mimic the feature of shallow layers [23], [33], [34], [39]. However, considering the increasing number of semantic segmentation tasks, these strategies could appear too restrictive, which will hurt the plasticity of the model and causes limited scalability. Therefore, we introduce the cosine similarity to align the high-frequency embedding consistencies between the shallow layer of the old and new models. Specifically, let us assume that the shallow layer θ_t^s of the new model Θ_t and the corresponding shallow layer θ_{t-1}^s of the old model Θ_{t-1}. The RCS loss can be defined as follows:

$$\mathcal{L}_{\text{rcs}}\left(x_i^t\right) = \frac{1}{W' \times H'} \sum_{m\in\mathcal{R}} 1 - \cos\left(a_m^t, a_m^{t-1}\right) \tag{7}$$

where a_m^{t-1} and a_m^t represent the mth feature embedding of feature maps $\theta_{t-1}^s(x_i^t)$ and $\theta_t^s(x_i^t)$, respectively, $\cos(\cdot)$ represents the cosine similarity operator, and $\mathcal{R} = \{1, 2, \ldots, W' \times H'\}$ represents all spatial positions of the feature map.

The RCS loss not only preserves the generalization of the learned tasks but also provides more flexibility to fit the unseen tasks. Note that the feature vectors are normalized to feature embeddings when calculating the RCS loss. As shown in Fig. 6, the embeddings all lie on a high-dimensional sphere. Therefore, our RCS loss encourages the angle of embeddings at the same spatial position to be close to each other. Compared with the constraints that directly freeze weights or mimic the feature of the old model, we only consider the orientation of features but not the magnitudes to protect the high-frequency stationary information. Therefore, our proposed RCS loss can avoid the rapid consumption of network capacity, leaving open access capacity to solve unseen tasks. This is a crucial consideration when learning on long sequences tasks. The

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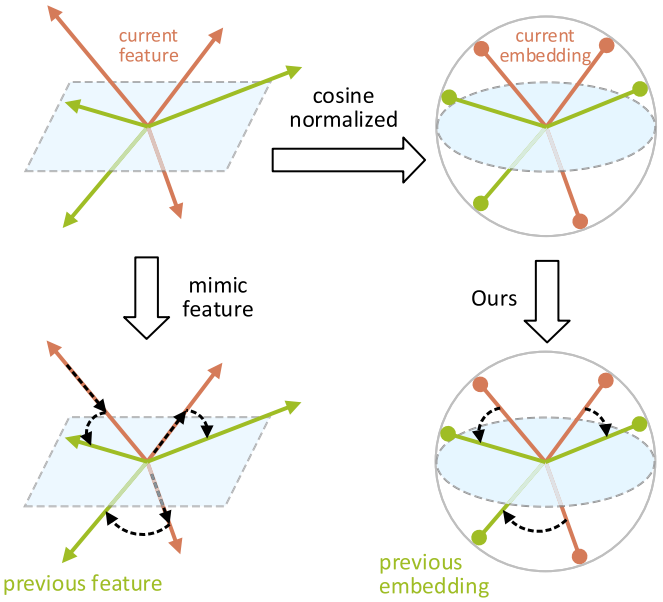

Fig. 6. Illustration of mimicking feature and RCS loss. The former simultaneously constrains the magnitudes and orientation of the features. Ours only consider the orientation of features, which can avoid the rapid consumption of network capacity.

qualitative analysis of high-frequency stationary information is given in Section IV-E2.

E. Integrated Optimization

Our approach is proposed to handle the catastrophic forgetting problem in task continual learning when the access to old semantic segmentation data is limited. Combining the losses presented above, we define the total loss during the learning step t as follows:

$$\mathcal{L}^t_{\text{total}} = \frac{1}{n_t}\sum_{i=1}^{n_t}\Bigg(\underbrace{\mathcal{L}_{\text{ce}}\left(x_i^t, y_i^t\right)}_{\text{new task } T_t} + \underbrace{\alpha\mathcal{L}_{\text{opd}}\left(x_i^t\right)+\beta\mathcal{L}_{\text{pas}}\left(x_i^t\right)+\gamma\,\mathcal{L}_{\text{rcs}}\left(x_i^t\right)}_{\text{old tasks } T_{1:t-1}}\Bigg) \tag{8}$$

where α, β, and γ denote the loss weights to balance the objective function. The cross entropy loss is applied to learn the new task T_t, and the other three loss function is applied to preserve the knowledge of old tasks $T_{1:t-1}$. The pipeline of our framework is shown in Algorithm 1. Although the continual learning network evolves during the continual process, there are no extra added parameters except the head classifiers. Besides, our framework does not need access to training samples of old tasks. Therefore, our method can be applied in practical applications with strict computational burden, memory cost, and storage budget.

Algorithm 1 Continual Learning With Structured Inheritance

Input:
$\mathcal{T} = (T_1, T_2, \ldots, T_t, \ldots)$: the sequence segmentation tasks;
$\mathcal{D}^t_{\text{train}} = \{x_i^t, y_i^t\}|_{i=1}^{n_t}$: training data of task T_t;
α, β, γ: weight parameters of different objects;
Output:
Θ_t: continual learning model optimized for tasks T_t;

while T_t **do**
  Get the training data $\mathcal{D}^t_{\text{train}}$ of current task;
  if $t = 1$ **then**
    Initialize Θ_1 with pretrained weights;
    Employ Θ_1 on training data $\mathcal{D}^1_{\text{train}}$: $f(x_i^1; \Theta_1)$;
    Get cross entropy loss $\mathcal{L}_{\text{ce}}(x_i^1, y_i^1)$ (3);
    Update the parameters of Θ_1:
      $\Theta_1 = \text{argmin}_{\{\theta_1^k\}}\mathcal{L}_{\text{ce}}$;
  else
    Load weights Θ_t from Θ_{t-1};
    Employ Θ_{t-1} on training data $\mathcal{D}^t_{\text{train}}$: $f(x_i^t; \Theta_{t-1})$;
    Employ Θ_t on training data $\mathcal{D}^t_{\text{train}}$: $f(x_i^t; \Theta_t)$;
    Get cross entropy loss $\mathcal{L}_{\text{ce}}(x_i^t, y_i^t)$ (3);
    Get OPD loss $\mathcal{L}_{\text{opd}}(x_i^t)$ between the output of Θ_{t-1} and Θ_t (4);
    Get PAS loss $\mathcal{L}_{\text{pas}}(x_i^t)$ between the end of shared layer $\theta^e_{t-1}(\cdot)$ and $\theta^e_t(\cdot)$ (6);
    Get RCS loss $\mathcal{L}_{\text{rcs}}(x_i^t)$ between the corresponding shallow layer $\theta^s_{t-1}(\cdot)$ and $\theta^s_t(\cdot)$ (7);
    Integrate the total loss $\mathcal{L}^t_{\text{Total}}$ (8);
    Update the parameters of Θ_t:
      $\Theta_t = \text{argmin}_{\{\theta_t^k\}}\mathcal{L}_{\text{total}}$;
  end if
end while
return Θ_t.

IV. Experiments

In this section, we first provide a detailed description of the datasets and experimental settings. Subsequently, we analyze and evaluate the effectiveness of our framework and compare it with state-of-the-art methods.

A. Datasets

We quantify the effectiveness of our approach on four remote sensing segmentation datasets, including the ISPRS semantic labeling contest dataset [40] composed of Potsdam and Vaihingen, DeepGlobe land cover challenge dataset [41], and GCSS [42], [43].

1) Potsdam: The Potsdam dataset is collected in the German city of Potsdam by airborne sensors, which contains 24 training images and 14 test images. The image size is 6000 × 6000 pixels and the spatial resolution is 5 cm. This dataset consists of four channels: near-infrared (NIR), red (R), green (G), and blue (B). These images are annotated into six classes: Building, Tree, Low Vegetation, Car, Impervious Surface, and Background. Following the prior works [56], we select seven images from training images as the validation set. Note that we do not use the digital surface model (DSM) or normalized DSM (nDSM) data.

2) Vaihingen: The Vaihingen dataset is collected in the German city of Vaihingen, which contains 16 training images and 17 test images. The size of these images is approximately

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TABLE I
STATISTICS OF THE FOUR SEMANTIC SEGMENTATION DATASETS

Data Sets	#Category	#Pixels	Resolution	Channels	Areas	Site	Source
Potsdam [40]	6	864,000,000	5 cm	NIR, R, G, B	Urban	Germany	airborne sensors
Vaihingen [40]	6	78,087,806	9 cm	NIR, R, G	Urban	Germany	airborne sensors
DeepGlobe [41]	7	4,812,141,312	50 cm	R, G, B	Rural	Thailand, Indonesia, India	DigitalGlobe +Vivid
GCSS [42], [43]	9	209,715,200	$<$ 1m	R, G, B	Urban, Rural	American	GF-2 satellite

2400 $\times$ 2000 pixels, and the spatial resolution is 9 cm. Each tile contains three channels corresponding to the NIR, R, and G domains. These images are densely annotated into six classes (with the same classes as for Potsdam). Following the prior works [57]–[59], we select five validation images from the training set. We do not use the DSM or nDSM data, too.

3) DeepGlobe: The DeepGlobe land cover dataset mainly focuses on rural areas in Thailand, Indonesia, and India. This dataset is sampled from the DigitalGlobe +Vivid and contains a total of 1146 RGB images, of which 803 have publicly available ground truth labels. The image size is 2448 $\times$ 2448 pixels, and the spatial resolution is 50 cm. Following the common setting [60], we split the 803 images into 455 training images, 142 validation images, and 206 testing images, respectively. These images are densely annotated into seven classes of landscape regions, including Urban land, Agriculture land, Rangeland, Forest land, Water, Barren land, and Unknown.

4) GCSS: The GCSS includes urban and rural areas in American, which is captured by the GF-2 satellite. It contains 800 RGB images of size 512 $\times$ 512 pixels with spatial resolution better than 1 m, 300 of these images are used as test images. The annotation contains nine classes, including Building, Road, Vehicle, Field, Water, Tree, Grass, Impervious surfaces, and Other.

As presented in Table I, the statistics show that there are significant differences between the above datasets. As shown in Fig. 1, the differences in factors such as resolution and image methods result in different distributions.

B. Implementation Details

1) Training Details: We adopt the PSPNet [53] with ResNet-50 [55] as the task shared layers of our continual learning network, the output stride of ResNet is set to 1/8, and it is initialized with ImageNet [61] pretrained model. The classifiers (one 1 $\times$ 1 convolution layer followed by the softmax operation) is treated as task-specific layer unless otherwise specified. The network is trained with stochastic gradient descent (SGD) with a weight decay of 0.001 and momentum of 0.9. We use the poly strategy to update the learning rate for each iteration. We set the base learning rate of the first learning step to 0.02 and followings to 0.001. Following the work [55], we use the liner warmup with iteration 500 and ratio 0.1 for each learning step to avoid instability in the initial training stage. All baselines and our methods use the same setting for the same dataset. Follow the prior works [24]–[26], [32], [33], [35], in order to avoid the loss of knowledge of the old tasks, we set the temperature τ of OPD loss to 1 and set the loss weight α in accordance with τ^2. The other two loss weights β and γ are set to maintain a balance with the distillation loss. We crop the images to 513 $\times$ 513 and set the batch size to 12. We train the model 10k iterations for Vaihingen, 50k iterations for Potsdam, 10k iterations for DeepGlobe, and 10k iterations for GCSS in each learning step. For data augmentation, we adopt the same policy of PSPNet. Specifically, we apply random horizontal flip, random resize, random rotation, and random Gaussian blur. For inference, consider the large size of the images, we use a sliding window strategy with overlapping stride 1/3 to verify performance on a single scale and original inputs. Due to the randomness of network training, all experiments were run three times with different random seeds, and the mean performance is reported. All experiments in this work are implemented by the Pytorch [62] framework. One NVIDIA Tesla V100 GPU with 32 GB memory is used in our experiments.

2) Evaluation Metrics: The mean intersection over union (mIoU), average continual accuracy (A), and average forgetting (F) [22] are introduced to evaluate the performance of our method. We use the mIoU score to fairly evaluate the performance on all sequence semantic segmentation tasks, let $\text{mIoU}_{k,j}$ be the accuracy calculated on the test set of the jth task ($j \leqslant k$) after training the continual learning network from task T_1 to T_k, which is defined as follows:

$$\text{mIoU}_{k,j} = \frac{1}{C_j} \sum_{x=1}^{C_j} \frac{p_{xx}}{\sum_{y=1}^{C_j} p_{xy} + \sum_{y=1}^{C_j} p_{yx} - p_{xx}} \tag{9}$$

where C_j denotes the number of categories of jth task, p_{xy} denotes the number of pixels of category x predicted to category y. Then, the average continual accuracy up to task T_k is defined as follows:

$$A_k = \frac{1}{k} \sum_{j=1}^{k} \text{mIoU}_{k,j} \tag{10}$$

note the higher of A_k is better. Moreover, the average forgetting is introduced to calculate the performance of the forgetting profile for different continual learning methods. Specifically, the average forgetting up to task T_k is defined as

$$F_k = \frac{1}{k-1} \sum_{j=1}^{k-1} f_j^k \tag{11}$$

here f_k^j is the forgetting of task T_j after the model has been continually trained to task T_k ($k > j$) and computed as

$$f_j^k = \max_{l \in \{1,\ldots,k-1\}} \text{mIoU}_{l,j} - \text{mIoU}_{k,j} \quad \forall j < k \tag{12}$$

the lower of F_k, the better of the method, which implies less forgetting on old tasks.

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TABLE II
PERFORMANCE COMPARISONS (MIOU) FOR CONTINUAL LEARNING TASK STARTING FROM DEEPGLOBE TO POTSDAM TO GCSS

Methods	step 1	step 2		step 3		
	DeepGlobe	DeepGlobe (old)→Potsdam (new)		DeepGlobe (old)→ Potsdam (old) → GCSS (new)		
Feature Extraction [63]	71.89	71.89 (ref)	38.26 (-40.36)	71.89 (ref)	38.26 (ref)	28.38 (-33.41)
Finetuning [64]	–	15.85 (-56.04)	78.62(ref)	9.38 (-62.51)	19.06 (-19.2)	61.79 (ref)
EWC [18]	–	19.23 (-52.66)	73.94 (-4.68)	16.43 (-55.46)	17.87 (-20.39)	56.35 (-5.44)
RW [22]	–	18.04 (-53.85)	75.60 (-3.02)	11.51 (-60.38)	22.04 (-16.22)	55.67 (-6.12)
PI [21]	–	17.72 (-54.17)	77.25 (-1.37)	10.95 (-60.94)	28.84 (-9.42)	58.21 (-3.58)
LWF [23]	–	61.80 (-10.09)	73.69 (-4.93)	54.55 (-17.34)	57.74 (+19.48)	48.45 (-13.34)
KD-BCE [32]	71.74	60.46 (-11.43)	77.77 (-0.85)	56.32 (-15.57)	67.24 (+28.98)	57.56 (-4.23)
ILT [33], [34]	–	67.52 (-4.37)	73.20 (-5.42)	55.59 (-16.3)	37.49 (-0.77)	55.75 (-6.04)
Ours	–	**70.12 (-1.77)**	**77.81 (-0.81)**	**68.57 (-3.32)**	**71.33 (+33.07)**	**58.90 (-2.89)**
Multi-task training [44]	–	69.00 (-2.89)	69.35 (-9.27)	63.28 (-8.61)	67.73 (+29.47)	53.74 (-8.05)

C. Compared Methods

We compare the proposed framework with several state-of-the-art methods to evaluate the effectiveness of avoiding catastrophic forgetting. Specifically, we reimplement the results of six different continual learning methods, three regularization methods, and three-parameter isolation methods.

For the first category, we choose LWF [23], KD-BCE [32], and incremental learning techniques (ILT) [33], [34]. LWF is the first work to introduce the KD strategy [31] to continual learning scenarios. Following the procedure of LWF, we first train the task-specific layer of the new task and then jointly train all parameters until convergence based on 4. KD-BCE replaces the cross entropy loss function to binary CLS for each class, even if the different classes are in the same label space. ILT is another method proposed to achieve continual learning in semantic segmentation, which proposed four KD variants. We utilize the feature alignment distillation loss as the implementation of ILT.

In the second category, we select EWC [18], PI [21], and RW [22], which are originally designed for image CLS. Specifically, EWC introduced the Fisher Information Matrix based on the analysis of sequential Bayesian estimation. PI utilized the PI on all old tasks. RW combined the PI with EWC to measure parameter importance. Note that we do not consider the network expansion methods. Because these methods are not suitable for practical applications with limited computational burden, memory cost, and storage budget.

Besides, we also compare three common methods, which provide the upper bound of the performance or reference accuracy. The first method is feature extraction [63], which freezes the task-shared layers and task-specific layers of old tasks and only optimizes the specific layer belonging to the new task. Feature extraction provides the reference accuracy for old tasks. The second method is finetuning [64], which trains the whole shared layers for the new task. Finetuning provides the reference accuracy for a new task. However, it will inevitably cause a dramatic accuracy drop on old tasks. The final method is multitask training [44], [45]. We introduce the multitask training strategy proposed from MT-DNN [45], which randomly selects a mini-batch from all training tasks and updated according to the task-specific objective. Note that it differs from the above-mentioned methods, where only the data of the current task are available. Multitask training can access the data for each task and optimize all tasks simultaneously, which provides the upper-bound for continual learning. Note that the training time of multitask training is often longer than that of continual learning methods.

D. Quantitative Analysis

1) Performance Comparisons: In this section, we first compare our method on a more realistic and complex multiple task continual learning scenario, that is, there are significant differences in the sequence semantic segmentation tasks. Then, we consider the effectiveness of our method in a similar task scenario.

a) Multiple task scenario: As shown in Table II, the first in which the continual learning network observes the dataset is DeepGlobe [41], the second where the network learns the dataset is Potsdam [40], and the third is GCSS dataset [42], [43]. The number of categories, resolution, dataset size, imaging method, and other information of the three datasets are different. As expected, the performance of Feature Extraction [63] on the new task drops sharply, because it freezes all parameters except the newly added parameters. Similarly, without employing any constraints, finetuning [64] performs poorly, completely forgetting the knowledge of old tasks. Although the starting-task DeepGlobe has far more pixels than Potsdam as shown in Table I, the network starts with a rich representation after learning the first task, still unavoidable catastrophic forgetting. Unexpectedly, the parameter isolation methods cannot avoid the catastrophic forgetting problem. Compared with finetuning, PI [21] can only bring a slight improvement to the performance of old tasks. Although RW [22] and EWC [18] performed slightly better on old tasks, they sacrifice the performance of the new task. We think this is because the assumption of parameter independence of such methods is too strong for semantic segmentation tasks. On the contrary, in learning step 2, the mIoU scores of the old task DeepGlobe obtained by LWF [23], KD-BCE [32], and ILT [33], [34] are 61.80%, 60.46%, and 67.52%, respectively. This proves the effectiveness of the KD strategy. However,

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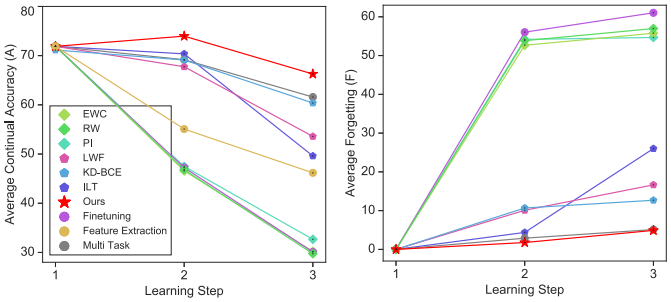
TABLE III

PERFORMANCE COMPARISONS (IOU) FOR CONTINUAL LEARNING TASK ON THE OLD TASK DEEPGLOBE

Methods	Urban	Agric.	Range.	Forest	Water	Barren	mIoU
Fea. Ext. [63]	78.54	87.82	41.17	80.72	80.72	62.37	71.89
Finetun. [64]	0.16	56.10	0.02	0.00	0.00	0.00	9.38
EWC [18]	8.83	46.32	3.71	19.46	19.22	1.06	16.43
RW [22]	1.96	31.61	2.13	16.65	2.16	14.53	11.51
PI [21]	0.22	56.93	0.05	7.69	0.00	0.80	10.95
LWF [23]	72.95	76.42	12.79	42.99	74.74	47.39	54.55
KD-BCE [32]	76.58	77.11	7.64	66.38	73.88	36.36	56.32
ILT [33]	57.60	77.28	15.08	64.83	72.81	45.94	55.59
Ours	**78.28**	**85.61**	**32.31**	**77.35**	**79.70**	**58.16**	**68.57**
Mul.-tas. [44]	72.22	82.42	24.62	74.83	77.62	47.99	63.28



(a) (b)

Fig. 7. Average performance of different methods for continual learning tasks starting from DeepGlobe to Potsdam to GCSS. (a) Average continual accuracy curve (higher is better). (b) Average forgetting curve (lower is better).

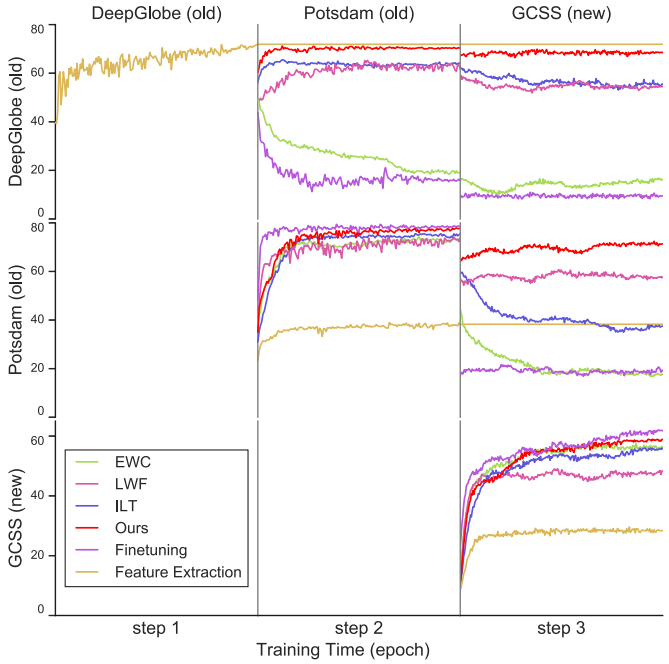


Fig. 8. Performance curves for different methods. We evaluate the performance on the test set each training epoch of each method for each task as the tasks are sequentially learned. Y-axis shows the mIoU score of different tasks as the training time increases. X-axis means the learning step. Note that learning step 1, step 2, step 3 can only observe the training samples of DeepGlobe, Potsdam, and GCSS, respectively. (Best viewed in color.)

when continuing to learn on the new task GCSS, these methods still cause performance degradation.

Compared with the methods mentioned above, our method can obtain a further boost whether it is old tasks or a new task. Specifically, in learning step 2, our method achieves the mIoU score of 77.81% for the new task Potsdam, which is only 0.81% lower than the reference score (Finetuning). For the old task DeepGlobe, our method gets the mIoU score 70.12%, which is also very close to the reference score (Feature Extraction). In learning step 3, our method can still retain the performance on old tasks while ensuring at the same time high accuracy on the new task. This implies that our proposed structured inheritance effectively solves the catastrophic forgetting problem of remote sensing semantic segmentation tasks.

As shown in Table III, we show the IoU accuracy for each class on the old task DeepGlobe after learning step 3. In the absence of access to the training data, our method successfully retains the knowledge of each category. As a comparison, others always result in a significant drop in IoU accuracy of some classes. Especially for the Rangeland class that is the hardest to classify, the mIoU score of ours is 19.52% higher than that of LWF. This proves that our method achieves good stability, thereby effectively avoiding catastrophic forgetting problems.

As shown in Fig. 7, the average continual accuracy curve and average forgetting curve of different methods show that our method has a better performance. Especially in learning step 3, our method can still maintain a stable level of performance. On the contrary, ILT proposed to mimic the features of the old model, resulting in the rapid decline of network capacity, thereby causing a sharp drop in performance. More importantly, we can find that our method outperforms the upper-bound (multitask training) [44]. Due to the huge difference between multiple tasks, although the multitask training access all training data, the performance on each task is not ideal. This indicates that the continual learning strategy can completely replace multitask training. Not only does it not require old training data, but the training time will also be greatly reduced.

Fig. 8 shows the performance curves plot each training epoch of different methods. Note that the multitask training is not shown in this figure because the training time is longer than others. In learning step 2, when adding the first new task Potsdam, all methods except Feature Extraction produce acceptable mIoU scores on Potsdam. Our method achieves performance similar to that of Finetuning. For the old task DeepGlobe, which cannot access the training data, EWC and Finetuning have encountered catastrophic forgetting. In learning step 3, when adding the second new task GCSS, our proposed method consistently performs well in overcoming forgetting. On the contrary, although the training data of the current task GCSS is accessible, the comparison methods still have performance degradation, and the performance degradation on the old tasks is more serious.

b) Similar task scenario: In this section, we select Vaihingen as the first task and Potsdam [40] as the second task

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TABLE IV

PERFORMANCE COMPARISONS (MIOU) FOR CONTINUAL LEARNING TASKS STARTING FROM VAIHINGEN TO POTSDAM

Methods	Vaihingen (old) → Potsdam (new)		A
Feature Extraction [63]	81.33 (ref)	54.34 (-25.48)	67.84
Finetuning [64]	31.36 (-49.97)	79.82 (ref)	55.59
EWC [18]	59.43 (-21.90)	78.38 (-1.44)	68.91
RW [22]	46.94 (-34.39)	79.05 (-0.77)	63.00
PI [21]	31.78 (-49.55)	**79.66 (-0.16)**	55.72
LWF [23]	78.86 (-2.47)	77.67 (-2.15)	78.27
KD-BCE [32]	79.29 (-2.04)	77.98 (-1.84)	78.64
ILT [33], [34]	80.26 (-1.07)	77.56 (-2.26)	78.91
Ours	**81.24 (-0.09)**	78.19 (-1.63)	**79.72**
Multi-task training [44]	77.92 (-3.41)	75.19 (-4.63)	76.56

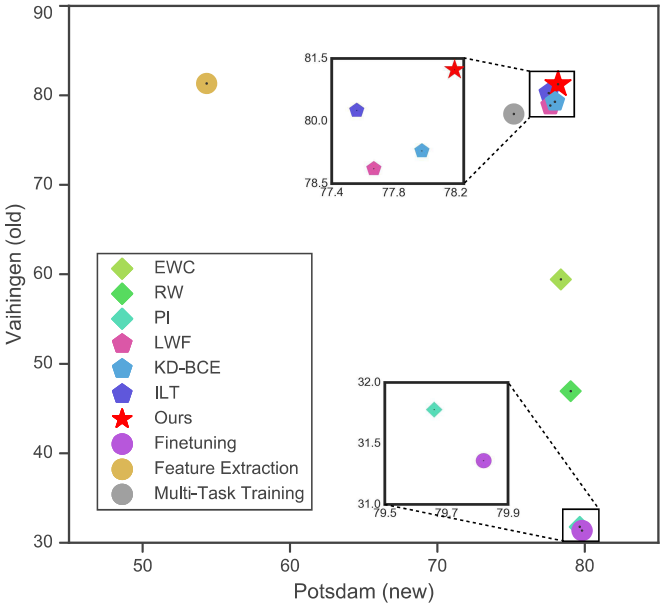

Fig. 9. Performance comparisons (mIoU) for different continual learning methods. The semantic segmentation task is from Vaihingen to Potsdam. The star is our proposed method, the pentagons are distillation-based methods, the diamonds are parameter isolation methods, the circles are the traditional common methods. (Best viewed in color.)

to explore the effect of task similarity. As shown in Table I, we can find that these two datasets all focus on the scene of the urban areas. They have the same semantic labels and similar resolution. This allows the continual learning methods to start with a similar feature space before learning the new task.

In Table IV and Fig. 9, we show the performance comparisons of this scenario. For the new task, all methods achieve accurate results similar to the reference performance (finetuning) [64]. This illustrates the role of task similarity on the new task. However, the comparative methods still have the catastrophic forgetting problem. The performance of finetuning on Vaihingen is more than 50% lower than the reference performance (feature extraction) [63]. Even multitask training [44] that can access all training data of Vaihingen and Potsdam, the performance of each task still has a certain degradation. The parameter isolation methods still show a stronger forgetting of the old task, losing knowledge

TABLE V

EMPIRICAL EVALUATION (MIOU) OF DIFFERENT COMPONENTS FOR CONTINUAL LEARNING TASKS STARTING FROM DEEPGLOBE TO POTSDAM. OPD-N: THE OPD LOSS FOR THE NEW TASK

OPD	OPD-N	PAS	RCS	DeepGlobe (old) → Potsdam (new)		A
√				66.64 ± 0.28	78.48 ± 0.24	72.56
√	√			63.61 ± 0.31	**79.28 ± 0.19**	71.45
√		√		69.05 ± 0.32	77.91 ± 0.20	73.48
√			√	68.92 ± 0.19	78.07 ± 0.28	73.50
√		√	√	**70.12 ± 0.12**	77.81 ± 0.32	**73.97**

TABLE VI

PERFORMANCE COMPARISON FOR DIFFERENT PIXEL STRUCTURE LOSS FOR CONTINUAL LEARNING TASKS STARTING FROM DEEPGLOBE TO POTSDAM. MIMIC: FEATURE ALIGNMENT BETWEEN OLD AND NEW MODELS. AT: ATTENTION MAP TRANSFER. LOCAL: THE LOCAL PIXEL STRUCTURE METHOD. SKPD METHOD

Methods	DeepGlobe (old) → Potsdam (new)		A
OPD [31]	66.64 (ref)	**78.48 (ref)**	72.56
OPD + Mimic [33], [34]	67.49 (+1.03)	76.49 (-1.99)	71.99
OPD + AT [25], [65]	64.64 (-2.00)	78.38 (-0.10)	71.51
OPD + LOCAL [66]	64.13 (-2.51)	77.78 (-0.70)	70.96
OPD + SKPD [67]	65.70 (-0.94)	78.09 (-0.39)	71.90
OPD + PAS	**69.05 (+2.41)**	77.91 (-0.57)	**73.48**

while learning a new task. The distillation-based methods work better when starting from a similar task, but the small change in the dataset still causes their accuracy deterioration. Our method gets mIoU score 81.24% on the old task, which is only 0.09% lower than feature extraction, and achieves the best average continual accuracy of 79.72%. As shown in Fig. 9, we can find that our method realizes the best tradeoff between the two sequence tasks. This once again proves the effectiveness of our proposed framework.

2) Ablation Experiments: To evaluate the effectiveness of our method, we conduct comprehensive ablation experiments for continual learning tasks starting DeepGlobe [41] to Potsdam [40]. The effectiveness of our method is provided in Table V. Table VI and Fig. 11 detail the effect of the two structure-preserving penalties. Table VII shows the effect of the number of the task-specific layers.

a) Effect of each component: Table V shows the contribution of different components. First, we conduct experiments on the accuracy of OPD loss [31], and its mIoU scores on DeepGlobe and Potsdam are 66.64% and 78.48%, respectively. Considering the effectiveness of the KD strategy, we also try to utilize the OPD loss on the new task. We use OPD-N to represent this training process. Specifically, an expert model is introduced to train the new task, and the output probability of the optimized expert model is used as the supervision information for the continual learning model to learn the new task. The second row shows the effect of applying OPD-N loss. The mIoU score of Potsdam is slightly improved by 0.8%, while the performance of DeepGlobe decreased by 3.03%. The average score of applying the OPD-N is lower than directly applying cross entropy loss. Moreover, the training process of the expert model increases the training time and training complexity. To sum up, we give up using OPD-N.

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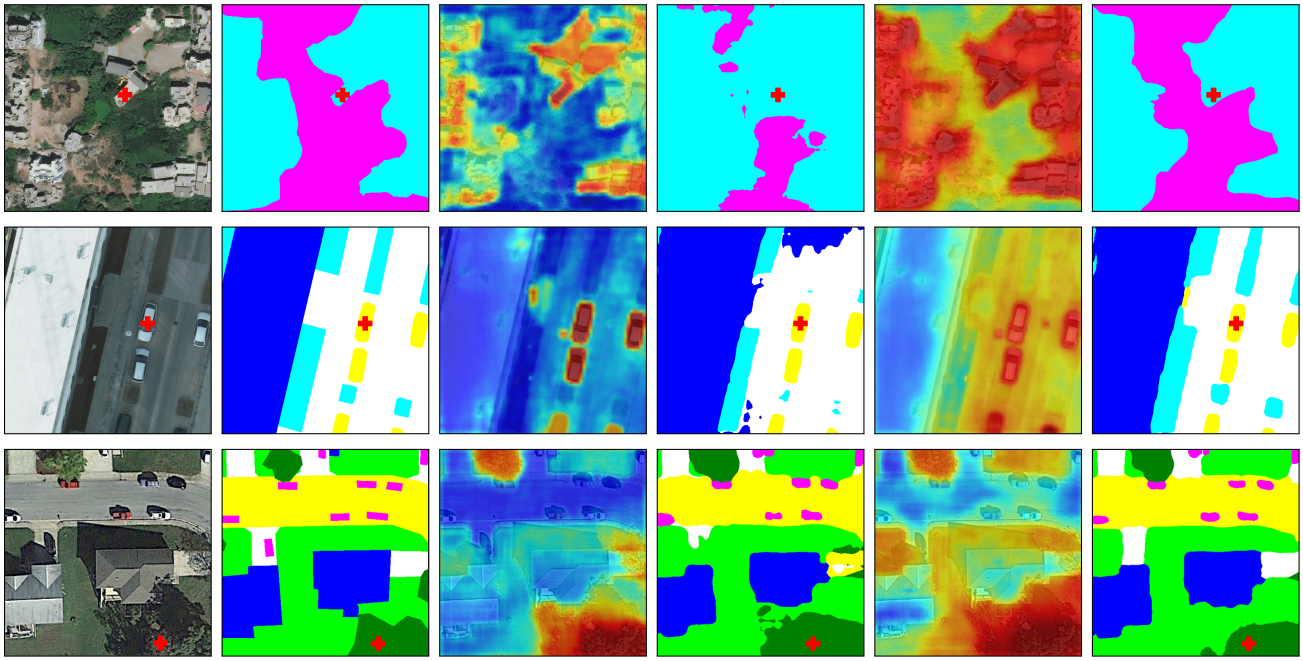

Fig. 10. Visualization of the pixel interactive information of a given pixel with and without the PAS loss. The pixel of *Urban land* from DeepGlobe (the first row), pixel of *Car* from Potsdam (the second row), and pixel of *Tree* from GCSS (the third row) are marked with a red cross. (a) Input image. (b) Ground truth. (c) Pixel affinity map without PAS. (d) Predict result without PAS. (e) Pixel affinity map with PAS. (f) Predict result with PAS. (Best viewed in color.)

Then, we evaluate the influence of our proposed structure-preserving penalties. The third row gives the performance of our model when PAS loss is used. The fourth row shows the result of using RCS loss. Compared with the model without using the structure-preserving penalty, they show improvements on old tasks of 2.41% and 2.28%, respectively. Finally, our method has all advanced developments after adding two structure-preserving penalties. The mIoU score on old task gains 3.48% than before. At meanwhile, the performance on the new task is still comparable. It verifies that the proposed two penalties are complementary. The experimental results prove the effectiveness of our proposed framework, which can learn new tasks while preserving high performance on old tasks.

b) Effect of PAS loss: In this section, we compare the variants of PAS loss: Mimic [33], [34], attention transfer (AT) [33], [34], LOCAL [66], and similarity preserving KD (SKPD) [66]. Mimic means aligning the features of each pixel directly between old and new models. AT is designed to make the new model transfer the attention maps of the old model. LOCAL considers the local PAS, which means the interaction between each pixel and the neighborhood pixels. The purpose of SKPD is to preserve the similarities between features of each class for image CLS. We replace our PAS loss with the above four loss functions to verify the effectiveness of our PAS loss.

As shown in Table VI, we can see that our PAS loss outperforms all the other methods. Note that only the average continual accuracy of ours is superior to the baseline. The superiority over Mimic and AT, which retain the knowledge for each individual pixel, is that we model the pixel interactive information other than aligning the feature for each pixel separately. Compared with LOCAL, it shows the effectiveness of modeling the whole pixel interactive information of the input image. Furthermore, compared with SKPD designed for image CLS, it shows that the pixel interactive information is more suitable for the dense prediction task. The visualization of pixel interactive information in Fig. 10 also proves the effectiveness of our proposed PAS loss.

c) Effect of RCS loss: To explore the effectiveness of our RCS loss and which layer is more suitable to protect the high-frequency stationary information, we conduct experiments on the four residual blocks of task shared layers and compare our method with the strategies of directly freezing weights or mimicking features [23], [33], [34], [39]. Fig. 11 shows the results. The circles from small to large correspond to L_1 to L_4 residual blocks of task shared layers, respectively.

Obviously, our proposed RCS loss achieves the best performance. As a comparison, freezing weights on shallow layers directly results in severe performance degradation on each task. Although the strategy of mimicking the feature brings performance gains on the old task, its best performance is still lower than our method. This verifies the effectiveness of RCS loss and further proves that the proposed representation structure loss can be used to protect the high-frequency stationary information.

As the depth of the selected layer increases, although the performance of our method on DeepGlobe (old task) is improving, the performance on Potsdam (new task) drops rapidly. This is because it reduces the network capacity available for the new task. Therefore, we select the L_1 in this work to apply the RCS loss. Furthermore, the strategies

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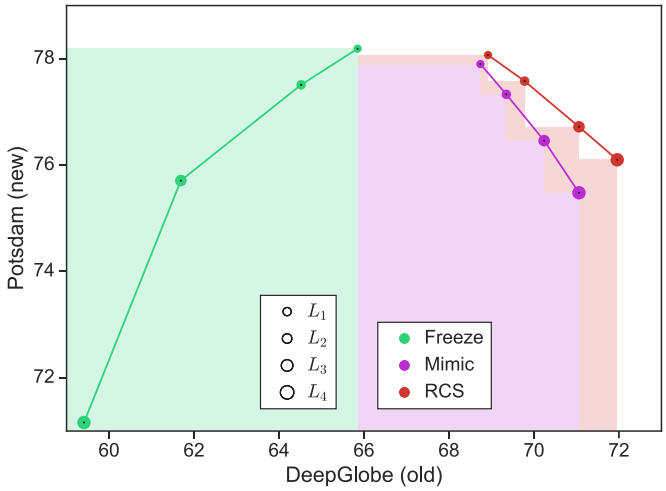

Fig. 11. Performance comparisons between our proposed RCS loss and the strategies of directly freezing weights or mimicking features. We evaluate the performance of adding them to the four residual blocks of task shared layers L_1 to L_4. The size of the black circle represents different residual blocks, the bigger the circle, the higher the layer. (Best viewed in color.)

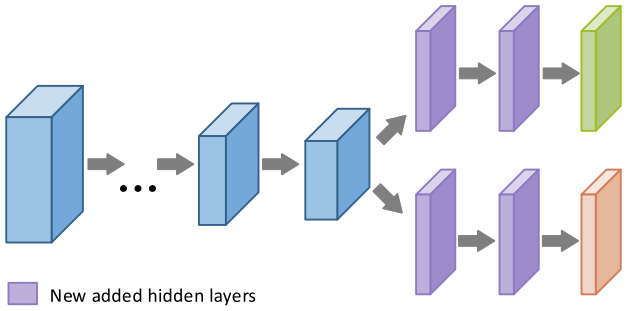

Fig. 12. Illustration for network modification. The more hidden layers are added to task-specific layers. (Best viewed in color.)

that directly freezing the weights or mimicking the features result in significant accuracy deterioration compared with our method. This proves that they consume more network capacity than ours, which will affect the learning of unseen tasks by continual learning network.

d) Effect of number of task-specific layers: In our work, the task-specific layer is a head classifier composed of one single 1×1 convolution layer. Considering that more task-specific layers may provide more advantages for each task, with more parameters to learn task-specific mapping rules. We explore whether adding the number of task-specific layers will bring performance gains. Specifically, we add the hidden layers between the end of task share layers and the task-specific layer for each task (see Fig. 12). All convolutions of the hidden layers are 3×3, the input and output channels are equal to the number of channels at the end of the task-shared layers.

As shown in Table VII, the results do not indicate any advantage to adding the number of task-specific layers. As the number of task-specific layers increases, performance on each task shows a trend of decline. Even in learning step 1 of normal supervised learning, more hidden layers lead to a decrease in

TABLE VII

PERFORMANCE COMPARISONS (MIOU) WHEN CHANGING THE NUMBER OF TASK-SPECIFIC HIDDEN LAYERS

#Hidden layers	step 1	step 2		A	F
	DeepGlobe	DeepGlobe (old)→Potsdam (new)			
0	**71.89 (ref)**	**70.12 (ref)**	**77.81 (ref)**	**73.97**	**1.77**
1	71.41 (-0.48)	67.87 (-2.25)	75.98 (-1.83)	71.93	3.54
2	70.86 (-1.03)	67.66 (-2.46)	64.44 (-13.37)	66.05	3.20
3	69.27 (-2.62)	67.50 (-2.62)	58.54 (-19.27)	63.02	**1.77**

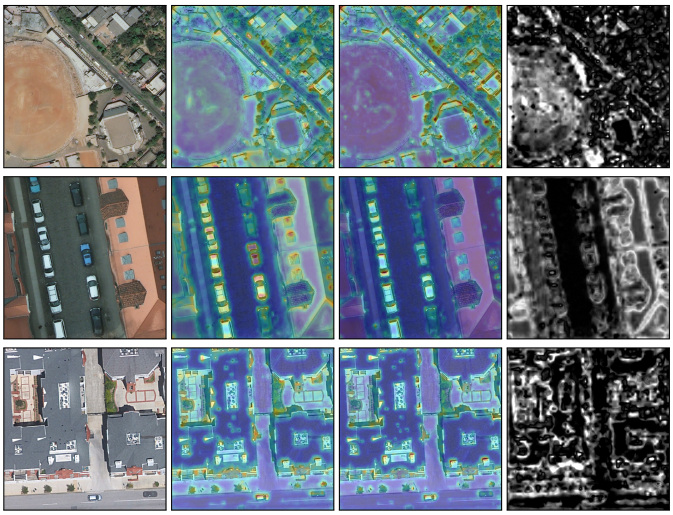

Fig. 13. Visualization of high-frequency stationary information with and without representation affinity structure loss. The images are from DeepGlobe (the first row), Potsdam (the Second row), and GCSS (the third row). (a) Input image. (b) Attention map without RCS. (c) Attention map with RCS. (d) Difference image between (b) and (c). (Best viewed in color.)

performance, especially when it is increased to three hidden layers, it shows a decrease of performance of 2.62%. The same is true for the performance on Potsdam in learning step 2. In addition, due to the need at the same time to preserve information about the old tasks, the performance degradation is even more severe. The mIoU score is decreased from 77.81% to 58.54%. Interestingly and quite intriguingly, there is only a little difference in the performance forgetting on DeepGlobe. On the one hand, it proves that increasing the number of task-specific layers has little effect on old task, and on the other hand, it shows that our framework is effective for dealing with catastrophic forgetting problem. Therefore, considering the experimental results and the additional storage brought by adding layers, we still regrade only the output nodes as task-specific layers.

E. Qualitative Analysis

1) Visualization of Pixel Interactive Information: As further qualitative analysis of PAS loss, we visualize the pixel interactive information with and without the PAS loss function (see Fig. 10). The input image in the first and second rows are from the old tasks DeepGlobe and Potsdam, respectively. The input image in the third row is from the new task Potsdam.

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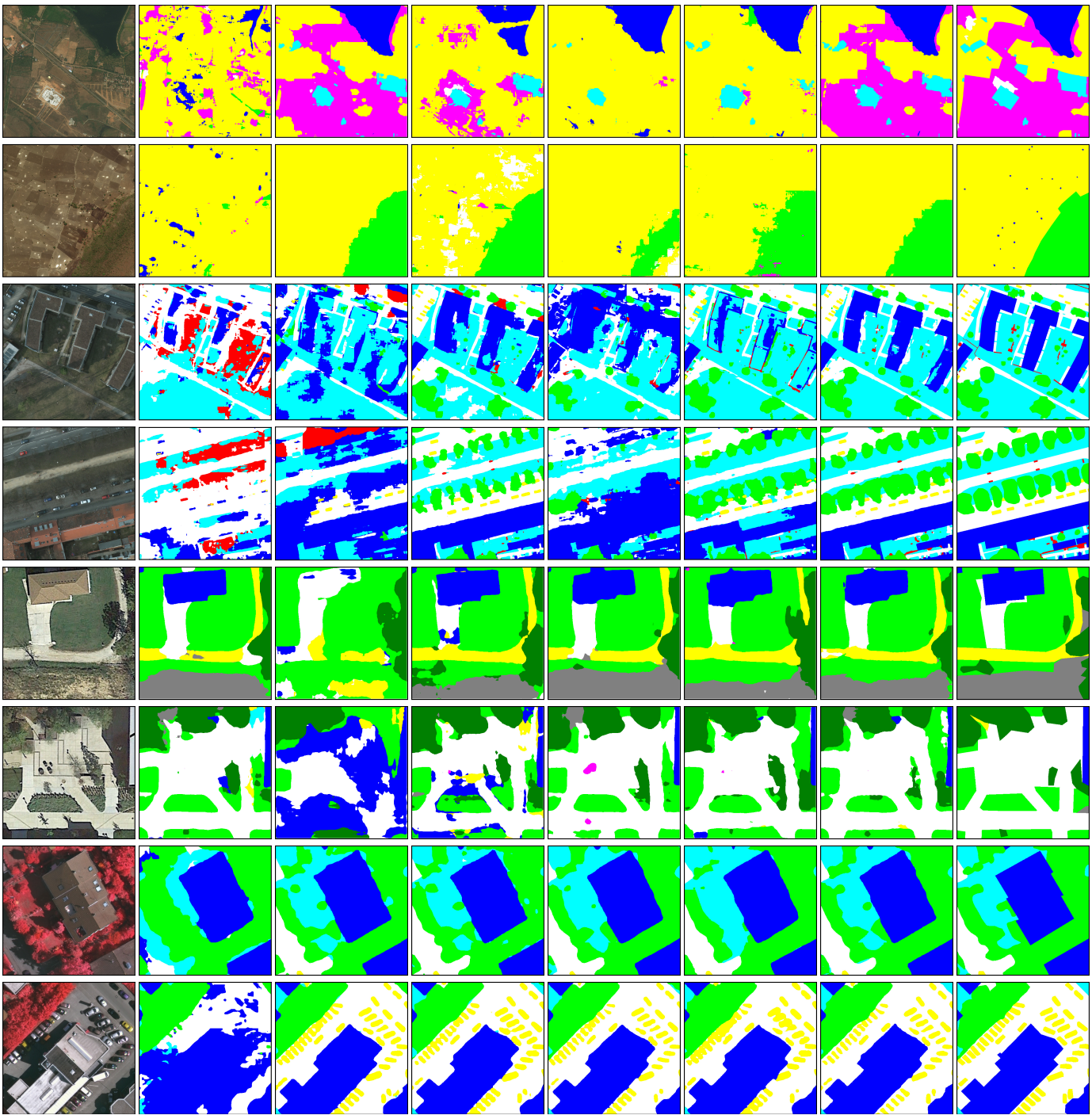

Fig. 14. Visualization of semantic segmentation results on DeepGlobe (the first two rows), Potsdam (the third and fourth row), GCSS (the third and fourth rows from the bottom), and Vaihingen (the bottom two rows). (a) Input image. (b) Finetuning [64]. (c) Feature Extraction [63]. (d) Multitask training [44]. (e) ILT [33]. (f) LWF [23]. (g) Ours. (h) Ground truth. Legend of DeepGlobe-Cyan: Urban land; Yellow: Agriculture land; Magenta: Rangeland; Green: Forest land; Blue: Water; White: Barren land; Black: Unknown. Legend of Potsdam and Vaihingen–Blue: Buildings; Green: Trees; Cyan: Low Vegetation; Yellow: Cars; White: Impervious surfaces; Red: Background. Legend of GCSS–Yellow: Road; Blue: Building; Dark Green: Tree; Green: Grass; gray: Field; Cyan: Water; Magenta: Vehicle; White: Impervious surfaces; Black: Other. Best viewed in color.

Due to the accessibility of the training data of the new task, the model without PAS loss can obtain the relatively pure pixel affinity map. However, it fails to capture the long-range dependence on old tasks, and only local similar patterns are highlighted. As a comparison, with the help of our proposed PAS loss, the long-range information is captured on old tasks and can be used to avoid further catastrophic forgetting. Besides, our proposed PAS loss can make the features between different classes more distinguishable. It can be seen that pixels belonging to different types have different responses with the selected pixels, which enhances the interclass variance and contributes to more robust predictions.

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2) Visualization of High-Frequency Stationary Information: To validate the effectiveness of our RCS loss, we visualize the high-frequency stationary information with and without RCS loss. We introduce the attention maps [65], which can highlight the most concerned information of the feature map. Specifically, it calculates the sum of the absolute values of the feature map across the channel dimension and then normalizes it to the range [0, 1]. As can be seen in Fig. 13, the attention maps of our method focus on the pixels belonging to the corners, textures, and edge, etc. Compared with the ground truth, the highlighted pixels are not affected by the semantic labels, which further proves that the shallow layer of the network learns the generalization and task-agnostic stationary features. As a comparison, the attention maps without RCS loss seem to be similar to ours. However, the difference images show the range of highlight pixels gradually expands, especially on the old tasks, which destroys the high-frequency stationary information.

3) Visualization of Segmentation Results: The qualitative results on the sequence tasks are presented in Fig. 14. We can find that finetuning [64] performs well on the new task GCSS. However, it suffers from catastrophic forgetting problems on the old tasks, and its segmentation results on DeepGlobe are almost like graffiti. On the contrary, feature extraction [63] successfully guarantees the performance on the initial task, but the segmentation results of the other tasks are poor. Although the training data of the new task can be accessed, it cannot make bricks without straw due to the limited network capacity. Multitask training [44] optimizes the training data of sequence tasks simultaneously, the difference in the feature space of the datasets can only compromise its optimization, resulting in suboptimal performance in all datasets. As for continual learning methods, ILT [33] and LWF [23] can only prevent catastrophic forgetting to a certain extent, and at the same time cause performance degradation on new tasks. Instead, our method achieves more accurate segmentation results in comparison with others, thanks to the proposed structure-preserving penalties, which successfully preserves pixel interactive information and high-frequency stationary information without prejudice to the new tasks.

V. Conclusion

In this work, we study the task of continual learning for semantic segmentation in aerial imagery. To overcome the catastrophic forgetting, we propose an effective and general framework based on structured inheritance. Specifically, we first present PAS loss to break the isolation of pixels and retain the pixel interactive information learned by the old tasks. Then, the RCS loss is introduced to protect the high-frequency stationary information. Extensive experiments prove the effectiveness of our proposed framework, which obtains superior performance over existing state-of-the-art continual learning methods. As for future work, we will consider applying our method on the longer sequences of semantic segmentation tasks or other dense prediction tasks, and try to consider a more general continual learning problem, that is, there is no knowledge about task identity and boundaries.

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